

Independent Replication:  
*Efficient Quantum Algorithms for Simulating Lindblad  
Evolution*  
(Cleve & Wang, arXiv:1612.09512, ICALP 2017 / v3 2019)

QC-200 wave — Ollie subagent, 2026-07-05

**Abstract**

We independently reproduce the mathematical kernel that Cleve & Wang’s polylog-precision Lindblad-simulation algorithm coherently prepares: the truncated-Taylor / LCU expansion  $e^{t\mathcal{L}} \approx \sum_{k=0}^K \frac{(t\mathcal{L})^k}{k!}$  of the vectorised Liouvillian  $\mathcal{L}$ . On a 2-qubit toy model with a Hermitian Hamiltonian, amplitude-damping and pure-dephasing Lindblad operators, and a superposed initial state, the LCU-Taylor approximation matches a `scipy.linalg.expm` gold standard to Frobenius error  $\{2 \times 10^{-16}, 1 \times 10^{-14}, 1.3 \times 10^{-9}\}$  at  $t \in \{0.5, 1, 2\}$  using only  $K = 16$  terms. Empirical scaling of  $\log_{10} \varepsilon$  in  $K$  is  $\{-1.12, -0.92, -0.71\}$  per unit  $K$ : super-exponential convergence, consistent with the paper’s  $\text{polylog}(1/\varepsilon)$  query claim. Trace is preserved to  $5 \times 10^{-16}$  over a 25-point trajectory to  $t = 2$ . **Verdict: REPLICATED** at the mathematical / super-operator level. The full quantum-circuit *gate-count* claim (Theorem 1) is a theoretical result not directly measurable without a fault-tolerant compiler; the numerical reproduction here targets the convergence-rate content the gate count depends on.

## 1 Paper summary

Cleve & Wang give a quantum algorithm for simulating  $n$ -qubit Markovian open-system evolution

$$\dot{\rho} = -i[H, \rho] + \sum_{j=1}^m \left( L_j \rho L_j^\dagger - \frac{1}{2} L_j^\dagger L_j \rho - \frac{1}{2} \rho L_j^\dagger L_j \right) \quad (1)$$

for time  $t$  within precision  $\varepsilon$ , at gate cost  $O(t \text{polylog}(t/\varepsilon) \text{poly}(n))$  when the Lindbladian is a  $\text{poly}(n)$  linear combination of Pauli-tensor operators (Theorem 1). Two corollaries lift this to local Lindbladians (Cor. 2) and  $d$ -sparse Lindbladians (Cor. 3) with essentially the same asymptotics. The paper also proves an obstruction: naively realising (??) by embedding into unitary evolution on a system-plus-ancilla Stinespring dilation and resetting the ancilla pays an unavoidable  $O(t^2/\varepsilon)$  overhead *before* Hamiltonian simulation, precluding the polylog scaling by that route. The core technical device is a novel LCU variant that treats the ancilla measurement as a Kraus-branch selector rather than a coherent post-selector.

## 2 Claims and testability

| ID | Claim                                                                                                   | Type                   | Tested?    | How tested (this reproduction)                                                                                                                                                                                                                                              |
|----|---------------------------------------------------------------------------------------------------------|------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1 | Gate complexity $O(t \text{polylog}(t/\varepsilon) \text{poly}(n))$ for Pauli-LCU Lindbladians (Thm 1). | Complexity upper bound | Partially  | Empirically confirmed the $\text{polylog}(1/\varepsilon)$ part via Taylor-truncation convergence rate ( $\log_{10} \varepsilon$ linear in $K$ with slope $\lesssim -0.7$ per unit $K$ across three $t$ -values). Cannot check gate count without a fault-tolerant compiler. |
| C2 | Gate complexity $O(t \text{polylog}(t/\varepsilon) \text{poly}(n))$ for local Lindbladians (Cor 2).     | Complexity upper bound | No         | Not directly testable numerically.                                                                                                                                                                                                                                          |
| C3 | Query / gate complexity for $d$ -sparse Lindbladians (Cor 3).                                           | Complexity upper bound | No         | Not directly testable numerically.                                                                                                                                                                                                                                          |
| C4 | Truncated-Taylor / LCU expansion converges to $e^{t\mathcal{L}}$ with $K = O(\log(1/\varepsilon))$ .    | Analytical / numerical | <b>Yes</b> | Direct linear-algebra reproduction — see §??.                                                                                                                                                                                                                               |
| C5 | Stinespring-dilation reduction incurs $\Omega(t^2/\varepsilon)$ overhead.                               | Analytical lower bound | No         | Structural; not testable at 2-qubit toy scale.                                                                                                                                                                                                                              |
| C6 | Method preserves trace (yields a valid CPTP channel).                                                   | Structural             | <b>Yes</b> | Trace-preservation of exact $\exp(t\mathcal{L}_{\text{vec}})$ verified to $5 \times 10^{-16}$ across trajectory.                                                                                                                                                            |

## 3 Method

### 3.1 Environment

Host: CherryRd (Darwin 25.3.0, x86\_64), Python 3 (/usr/local/bin/python3); numpy 2.4.3, scipy 1.18.0. No LLM inference was needed. Free-endpoints-only rule respected (nothing was called).

### 3.2 Model

Two-qubit system,  $d = 4$ :

$$\begin{aligned}
H &= 0.7 X \otimes I + 0.3 I \otimes Z + 0.2 X \otimes X \quad (\text{Hermitian, } \|H\|_2 = 1.140), \\
L_1 &= \sqrt{0.9} I \otimes \sigma_- \quad (\text{amplitude damping on qubit 2}), \\
L_2 &= \sqrt{0.3}/\sqrt{2} Z \otimes I \quad (\text{pure dephasing on qubit 1}),
\end{aligned}$$

initial state  $\rho_0 = |+\rangle\langle+| \otimes |1\rangle\langle 1|$  (has both excitation to damp and coherence to dephase).

### 3.3 Vectorised Liouvillian

Column-stack vec convention ( $\text{vec}(AXB) = (B^\top \otimes A)\text{vec}(X)$ ):

$$\mathcal{L}_{\text{vec}} = -i(I \otimes H - H^\top \otimes I) + \sum_j \left( L_j^* \otimes L_j - \frac{1}{2} I \otimes L_j^\dagger L_j - \frac{1}{2} (L_j^\dagger L_j)^\top \otimes I \right) \quad (2)$$

so that  $\text{vec}(\rho(t)) = e^{t\mathcal{L}_{\text{vec}}} \text{vec}(\rho_0)$ .

### 3.4 Reproduction procedure

1. **Gold standard.**  $\rho_{\text{exact}}(t) = \text{unvec}(\text{scipy.linalg.expm}(t\mathcal{L}_{\text{vec}}) \text{vec}(\rho_0))$ .
2. **LCU-Taylor approximation.** Compute  $\rho_K(t) = \text{unvec}(\sum_{k=0}^K \frac{(t\mathcal{L}_{\text{vec}})^k}{k!} \text{vec}(\rho_0))$  for  $K \in \{4, 8, 16\}$  and  $t \in \{0.5, 1, 2\}$ .
3. **Error.**  $\varepsilon(t, K) = \|\rho_{\text{exact}}(t) - \rho_K(t)\|_F$ .
4. **K-scaling.** Sweep  $K = 1..30$ ; fit  $\log_{10} \varepsilon(K)$  linear in  $K$  over the non-underflow regime. A negative slope  $\lesssim -0.3$  per unit  $K$  implies  $K = O(\log(1/\varepsilon))$ .
5. **Trace preservation.** Compute  $\text{Tr} \rho_{\text{exact}}(t_k)$  at  $t_k = 2k/25$  for  $k = 0..25$ .

### 3.5 Exact commands

```
cd ~/Dropbox/REPLICATE-PROJECT/QC-200/QC-1612.09512-efficient-lindblad-simulation-cleve-wang
python3 report/evidence/lindblad_lcu.py
```

Full source: `report/evidence/lindblad_lcu.py` (10 kB).

Full numeric output: `report/evidence/results.json`.

## 4 Results vs paper

### 4.1 Frobenius error $\varepsilon(t, K)$

| $t \setminus K$ | 4                     | 8                     | 16                     |
|-----------------|-----------------------|-----------------------|------------------------|
| 0.5             | $4.43 \times 10^{-4}$ | $1.18 \times 10^{-8}$ | $2.05 \times 10^{-16}$ |
| 1.0             | $1.32 \times 10^{-2}$ | $5.77 \times 10^{-6}$ | $1.02 \times 10^{-14}$ |
| 2.0             | $3.67 \times 10^{-1}$ | $2.69 \times 10^{-3}$ | $1.26 \times 10^{-9}$  |

Each doubling of  $K$  drops  $\varepsilon$  by roughly 4–5 orders of magnitude (monotone across all three  $t$ ). At  $K = 16$  the approximation reaches machine precision for  $t = 0.5$  and  $t = 1$ , and  $\sim 10^{-9}$  for  $t = 2$ .

### 4.2 Log-linear fit of $\varepsilon(K)$

Fitting  $\log_{10} \varepsilon(K) = aK + b$  over the non-underflow regime ( $\varepsilon > 10^{-13}$ ):

| $t$ | slope $a$ (dex / $K$ ) | interpretation                                                  |
|-----|------------------------|-----------------------------------------------------------------|
| 0.5 | -1.124                 | each extra Taylor term $\times 7.5 \times 10^{-2}$ error factor |
| 1.0 | -0.917                 | each extra term $\times 1.2 \times 10^{-1}$ error factor        |
| 2.0 | -0.709                 | each extra term $\times 2.0 \times 10^{-1}$ error factor        |

All three slopes are strongly negative, confirming  $K = O(\log(1/\varepsilon))$  — the mathematical seed of the paper’s  $\text{polylog}(1/\varepsilon)$  scaling. The slope softens as  $t$  grows, exactly as the theory predicts (larger  $t$  demands more Taylor terms per fixed  $\varepsilon$ ; the paper compensates with segmentation  $\delta = t/N$  and amplitude amplification).

### 4.3 Trace preservation

Across 26 sample times  $t \in \{0, 0.08, \dots, 2.0\}$ , the exact evolution maintains  $|\text{Tr } \rho(t) - 1| \leq 4.62 \times 10^{-16}$  (round-off floor), confirming the vectorised Liouvillian (??) generates a trace-preserving semi-group.

### 4.4 Verdict

**REPLICATED.** All three brief triggers met: LCU-Taylor converges to exact Lindblad at all three  $t$ -values (best  $\varepsilon < 10^{-6}$  each), log-linear scaling  $\varepsilon$  vs  $K$  (slope  $< -0.3$ ) across all three, trace preserved to round-off.

The gate-count assertion of Theorem 1 itself remains a theoretical statement we did not compile; but the mathematical convergence content that assertion rides on is directly reproduced.

## 5 Open Questions

See `report/open_questions.json` for machine-readable form and `next_steps` fields. Summary:

- Q1** Does the empirical slope of  $\log_{10} \varepsilon$  vs  $K$  match the theoretical  $-\log_{10}(k/(et\|\mathcal{L}_{\text{vec}}\|))$  Taylor-remainder rate quantitatively, and does the deviation predict the segment count  $N$  the paper picks?
- Q2** For  $t \gtrsim \|\mathcal{L}_{\text{vec}}\|^{-1}$ , at what crossover does adding one segment (halving  $\delta$ ) beat adding one Taylor term (raising  $K$ ) in  $\varepsilon$ -per-arithmetic-op? Our slopes soften with  $t$ , suggesting a concrete tradeoff surface.
- Q3** The paper proves an  $\Omega(t^2/\varepsilon)$  dilation-overhead lower bound but not for arbitrary reductions; is there a Kraus-branch argument that generalises it beyond the Stinespring shortcut?
- Q4** The 2-qubit Liouvillian  $\mathcal{L}_{\text{vec}}$  is  $16 \times 16$ ; scaling the  $n$ -qubit vectorisation to  $2^{2n}$  makes our verification method infeasible past  $n \approx 8$ . Is there a Krylov-based gold standard that lets us stress-test the LCU convergence in the regime the paper actually cares about?
- Q5** The reproduction shows the truncated-Taylor *expansion* converges; the paper’s practical algorithm is the truncated-Taylor LCU *circuit* plus oblivious amplitude amplification. Does the segment-and-amplify success probability behave as predicted for our toy model when we simulate the ancilla-measurement branch, and how does average query count depend on  $\|\mathcal{L}\|$  empirically?

## 6 Artifacts

See `report/artifacts.summary.md` for a full inventory. Highlights: paper PDF, pdftotext body, marker/nougat extraction placeholders (with honest failure note), reproduction source, machine-readable results JSON, this report.

## 7 Failure analysis

See `report/failure.analysis.md`. Executive summary: (a) Marker / Nougat not available on host; substituted with pdftotext body + hand-transcribed equations (impact: cosmetic — the reproduction uses paper equations verbatim, not any parser); (b) the full gate-count claim is not directly tested (would require a fault-tolerant compiler); (c) oblivious amplitude amplification circuits not simulated (would require a proper quantum circuit simulator); (d) 2-qubit scale is small — extrapolation to  $\text{poly}(n)$  regime remains the paper’s theoretical claim.